

A Gram-certificate proof that

$$\sigma_1(\mathbb{R}^2) \leq 0.6934$$

What is formalized in `Besicovitchs-1-2`, and why the constant is not sharp

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Abstract

This note describes the theorem verified in the Lean 4 development `Besicovitchs-1-2`: the planar one-dimensional rectifiability threshold satisfies $\sigma_1(\mathbb{R}^2) \leq 6934/10000$, improving the published bound of 0.7. The proof combines the six-point failure tree of the two-colour finite problem with thirty local Gram certificates. We explain the argument, state precisely what the machine checks, and since this is the natural question explain why the formalized constant is 0.6934 rather than the sharp six-point endpoint $s_* = 0.69330642\dots$, and what it would take to close the remaining gap of $9.4 \cdot 10^{-5}$.

1 The result

For a Borel set $E \subset \mathbb{R}^2$ with $\mathcal{H}^1(E) < \infty$, the lower density at x is

$$\Theta_*^1(E, x) = \liminf_{r \downarrow 0} \frac{\mathcal{H}^1(E \cap B(x, r))}{2r},$$

and $\sigma_1(\mathbb{R}^2)$ is the least threshold β such that $\Theta_*^1 \geq \beta$ almost everywhere forces countable 1-rectifiability. Besicovitch's 1/2 conjecture asserts $\sigma_1(\mathbb{R}^2) = 1/2$. The verified statement is

Theorem 1.1 (`Besicovitch.sigma_one_plane_le_6934_div_10000`).

$$\sigma_1(\mathbb{R}^2) \leq \frac{6934}{10000} = 0.6934.$$

The proof term depends on exactly the three axioms `propext`, `Classical.choice` and `Quot.sound`. It uses no `native_decide`, no `Lean.ofReduceBool`, no floating-point value, no hash, and no custom axiom.

2 Why the constant is not sharp

The six-point finite problem this development solves has an exact optimal constant,

$$\theta_6 = s_* = 0.693306421825904872690678414403710951\dots,$$

an algebraic number pinned by an isolated pair of radical equations. So a natural target is $\sigma_1(\mathbb{R}^2) \leq s_*$, and an earlier form of this repository pursued exactly that. The reason the formalized statement

is the weaker rational bound is not that the mathematics fails at s_* ; it is that *tightness destroys the certificates*.

At s_* the extremal configuration is attained: a half-turn pair of chords at which three structurally different packing mechanisms tie *simultaneously*. The weighted score that the argument must bound above by zero therefore has value exactly 0 at an interior point of the domain, with vanishing gradient. Any certificate that establishes a non-strict bound at a degenerate maximum must resolve the geometry near that point to the precision of the degeneracy. Concretely, in the earlier Bernstein formulation the margin available away from the extremiser was about 10^{-8} , and the adaptive subdivision needed to certify the neighbourhood of the tie grew until one of the ten coordinate charts admitted no finite tree at all: on that chart the score is identically 0 at the extremiser, so no strict Positivstellensatz certificate of the assumed shape exists.

Moving the chord to the rational value

$$c = \frac{3467}{2500} = 1.3868, \quad s = \frac{c}{2} = \frac{6934}{10000},$$

breaks every tie at once. The three mechanisms no longer agree, the extremal configuration is no longer feasible, and the weighted score acquires a uniform negative margin of about $5.8 \cdot 10^{-4}$ roughly four orders of magnitude more room than at s_* . That margin is what makes a *small, fixed* certificate family sufficient: thirty rectangles suffice where the sharp problem needed an unbounded adaptive tree.

The cost of this trade is quantified exactly: the gap between what is proved and the sharp six-point constant is

$$\frac{6934}{10000} - s_* = 9.357817 \cdot 10^{-5}.$$

The gap to the conjecture itself is of course far larger; $\sigma_1(\mathbb{R}^2) = 1/2$ remains open, and the six-point method cannot reach below θ_6 no matter how it is certified. A companion theorem in `Solution.lean`, `sStar_le_6934_div_10000`, records that the exact endpoint does lie below the rational target, so the two statements are consistent and the formalized bound is genuinely weaker only by the amount displayed above.

There is a second, upper constraint on the choice, which is why the constant is 0.6934 and not, say, 0.699. The routing separators that prune the case tree are themselves inequalities in the chord, and they cease to hold once c exceeds roughly 1.386850. The rational chord above sits just under that ceiling while clearing s_* from above, so the interval of admissible targets is narrow, and 6934/10000 is a natural point inside it.

3 Input from the six-point failure tree

Let e, p_1, p_2, w_1, w_2 lie in a real inner-product space with

$$\|e\| = 1, \quad \|p_i\| \leq 1, \quad \|w_i\| \leq 1,$$

and suppose the two sibling pairs are separated,

$$c \leq \|p_1 - p_2\|, \quad c \leq \|w_1 - w_2\|. \tag{1}$$

The finite failure tree retained unchanged from the sharp development reduces the failure of all nine packings at a matched sibling endpoint to three scalar inequalities

$$q_1 \geq 0, \quad q_2 > 0, \quad q_3 > 0, \tag{2}$$

in Lean `firstActiveFailureSlack_nonneg`, `secondActiveFailureSlack_pos` and `thirdActiveFailureSlack_pos`

The weighted pair score is, for weights $\lambda, \mu > 0$,

$$\begin{aligned}\mathcal{W}_{c,\lambda,\mu} = & (1 + \lambda)\|e - p_1 - w_1\| + \|e - p_2 - w_2\| \\ & + \frac{\mu}{2}(\|e - p_1\| + \|e - w_1\| + \|e - p_1 - w_2\| + \|e - w_1 - p_2\|) \\ & - \frac{(c-1)(\lambda/2+\mu)}{2}(\|p_1\| + \|w_1\|) - \frac{(c+1)\lambda/2+3c\mu}{2}(\|p_2\| + \|w_2\|) \\ & - \left(2c(2c-1) + \frac{\lambda}{2}(3c^2 - 3c + 2) + \mu(c^2 - c)\right),\end{aligned}$$

and a purely algebraic identity gives $q_1 + \lambda q_2 + \mu q_3 = \mathcal{W}_{c,\lambda,\mu}$ for *every* λ, μ (`weightedPairScore_configuration_eq`). Hence (2) forces $\mathcal{W}_{c,\lambda,\mu} > 0$.

The contradiction we must produce is an upper bound $\mathcal{W} < 0$. In the sharp development the weights were the Cramer weights of the endpoint system, algebraic numbers of high degree. Because the identity holds for arbitrary weights, we are free to pick

$$\lambda = \frac{1}{12}, \quad \mu = \frac{13}{14}, \quad (3)$$

which is what makes every subsequent coefficient a small rational. This substitution alone removes the entire exact-endpoint apparatus from the dependency path of the theorem.

4 One local Gram certificate

Fix a rectangle of second-child radii

$$l_P \leq \|p_2\| \leq u_P, \quad l_W \leq \|w_2\| \leq u_W. \quad (4)$$

The triangle inequality and (1) give $c - u_P \leq \|p_1\| \leq 1$ and $c - u_W \leq \|w_1\| \leq 1$; note that the lower bound uses u_P , not l_P .

Write $v_0 = e$, $v_1 = p_1$, $v_2 = p_2$, $v_3 = w_1$, $v_4 = w_2$. With the weights (3) the six positively weighted terms of $\mathcal{W}$ are $\|\xi_t\|$ for $\xi_t = \sum_j A_{tj}v_j$,

$$A = \begin{pmatrix} 1 & -1 & 0 & -1 & 0 \\ 1 & 0 & -1 & 0 & -1 \\ 1 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 & -1 \\ 1 & 0 & -1 & -1 & 0 \end{pmatrix}, \quad \omega = \left(\frac{13}{12}, 1, \frac{13}{28}, \frac{13}{28}, \frac{13}{28}, \frac{13}{28}\right).$$

Three elementary inequalities linearize the problem. For $\alpha_t > 0$ the quadratic tangent $\omega_t \|\xi_t\| \leq \alpha_t \|\xi_t\|^2 + \omega_t^2 / (4\alpha_t)$ replaces each norm by a quadratic form; the radial secant $-dr \leq -\frac{d}{l+u}r^2 - \frac{dl u}{l+u}$ (valid for $0 \leq l \leq r \leq u$) replaces each of the four radial penalties; and for $\eta_P, \eta_W \geq 0$ the separation multipliers $\eta_P(\|p_1 - p_2\|^2 - c^2) \geq 0$ and $\eta_W(\|w_1 - w_2\|^2 - c^2) \geq 0$ may be added freely. Expanding the squared norms,

$$\mathcal{W} \leq C + \sum_{i=0}^4 D_i \|v_i\|^2 + 2 \sum_{0 \leq i < j \leq 4} Q_{ij} \langle v_i, v_j \rangle, \quad (5)$$

with C, D, Q explicit rational functions of the certificate parameters and the box.

It remains to remove the inner products. A certificate carries a rational 3×5 matrix F ; put $B = F^\top F$, let $\rho_{ij} = -Q_{ij} - B_{ij}$, and let $\varepsilon_{ij} = \text{sign } \rho_{ij}$. The matrix

$$M = B + \sum_{i < j} |\rho_{ij}| (e_i + \varepsilon_{ij} e_j)(e_i + \varepsilon_{ij} e_j)^\top \quad (6)$$

is a sum of rank-one positive semidefinite matrices, hence positive semidefinite, and by construction $M_{ij} = -Q_{ij}$ off the diagonal. Since the Gram matrix $G = (\langle v_i, v_j \rangle)$ is also positive semidefinite, the Schur product theorem gives $\sum_{i,j} M_{ij} \langle v_i, v_j \rangle \geq 0$, that is

$$0 \leq \sum_i m_i \|v_i\|^2 - 2 \sum_{i < j} Q_{ij} \langle v_i, v_j \rangle, \quad m_i = B_{ii} + \sum_{j \neq i} |\rho_{ij}|.$$

Adding this to (5) cancels every inner product and leaves $\mathcal{W} \leq C + \sum_i \beta_i \|v_i\|^2$ with $\beta_i = D_i + m_i$. Finally each $\beta_i \|v_i\|^2$ is bounded on its radius interval by $T(\beta; l, u) = (\beta)_+ u^2 - (\beta)_- l^2$, giving a single rational number

$$U = C + \beta_0 + T(\beta_1; c - u_P, 1) + T(\beta_2; l_P, u_P) + T(\beta_3; c - u_W, 1) + T(\beta_4; l_W, u_W). \quad (7)$$

Proposition 4.1 (`weightedPairScore_le_of_gramCertificate`). *If $\alpha_t > 0$, $\eta_P, \eta_W \geq 0$ and $U \leq -\frac{1}{2000}$, then every configuration satisfying (1) and (4) in the unit ball has $\mathcal{W} \leq -\frac{1}{2000} < 0$.*

No eigenvalue computation, numerical positive-definiteness test, or polynomial arithmetic occurs anywhere: positive semidefiniteness is established *identically* from the rank-one decomposition (6).

5 The finite cover

Every feasible second-child radius lies in $[c - 1, 1]$. Seven intervals cover that range,

$$\begin{aligned} I_0 &= [c - 1, \tfrac{1}{2}], & I_1 &= [\tfrac{1}{2}, \tfrac{3}{5}], & I_2 &= [\tfrac{3}{5}, \tfrac{7}{10}], \\ I_3 &= [\tfrac{7}{10}, \tfrac{3}{4}], & I_4 &= [\tfrac{3}{4}, \tfrac{4}{5}], & I_5 &= [\tfrac{4}{5}, \tfrac{9}{10}], \\ I_6 &= [\tfrac{9}{10}, 1]. \end{aligned}$$

The score is symmetric under exchanging the two sibling pairs, so only the 28 ordered pairs $I_i \times I_j$ with $i \leq j$ are needed. The single square $I_2 \times I_2$ carries too little margin and is split once more using $I_{2a} = [\frac{3}{5}, \frac{13}{20}]$ and $I_{2b} = [\frac{13}{20}, \frac{7}{10}]$, contributing three rectangles instead of one. The total is $28 - 1 + 3 = 30$.

Theorem 5.1 (`weightedGeometricBound_gram`). *For every e, p_1, p_2, w_1, w_2 in a real inner-product space with $\|e\| = 1$, $\|p_i\|, \|w_i\| \leq 1$ and (1),*

$$\mathcal{W}_{c,1/12,13/14} \leq -\frac{1}{2000}.$$

Combined with the failure tree this contradicts $\mathcal{W} > 0$, so the six-point finite property holds at $s = 6934/10000$; the already formalized six-point-to-BPC and BPC-to-rectifiability transfers then give the theorem.

The binding certificate is the one on $I_3 \times I_5$, whose exact upper bound is

$$U = -\frac{141718938311359411043}{245112420775950000000000} = -0.000578179 \dots < -\frac{1}{2000},$$

so the whole family clears the uniform threshold with about 16% to spare.

6 The lower bound

The registry review observed that Lean’s real infimum returns 0 on the empty set, so an upper bound on σ_1 could in principle hold vacuously. The formalization therefore also proves the classical lower bound

$$\frac{1}{2} \leq \sigma_1(\mathbb{R}^2),$$

which pins the threshold to $[\frac{1}{2}, 0.6934]$ and excludes the vacuous case outright. The statement `one_half_le_sigma_one_plane` is compared alongside the upper bound.

A lower bound needs a set of finite positive length that is not countably rectifiable and yet has lower density at least $\frac{1}{2}$ almost everywhere. Besicovitch proposed one in 1938 and Dickinson proved its properties in 1939; we follow Capdevila’s 2026 account. Let

$$\alpha_n = 2^{-n^2}, \quad \beta_n = \alpha_n/n, \quad g = \sum_{n \geq 1} f_n,$$

where f_n is the square wave of period $2\alpha_n$ and amplitude β_n , and let Π be the graph of g over $[0, 1]$. Two estimates carry everything. Inside a level- n cell $[i\alpha_n, (i+1)\alpha_n)$ the function varies by at most $4\alpha_{n+1}/(n+1)$; across a level- n cell boundary it jumps by at least α_n/n . Mathlib contains no notion of rectifiability and no purely unrectifiable set, so the example and its three properties are built from scratch.

Measure. The first-coordinate projection is 1-Lipschitz, so $\mathcal{H}^1(\Pi(A)) \geq |A|$ for every $A \subset [0, 1]$; covering the graph over a cell by a cylinder of diameter at most $2\alpha_n$ gives $\mathcal{H}^1(\Pi(A)) \leq 2|A|$, first on intervals, then on open sets through the cells they contain, then on arbitrary sets by outer regularity. The sharp constant 1 is never needed. In particular the graph over a null set is null.

Density. At a point $x \in (0, 1)$ and a small radius r , choose the level n with $\alpha_{n+1} < r \leq \alpha_n$. The graph over the part of x ’s cell within θr of x lies in the ball of radius r about $(x, g(x))$, because the vertical variation inside the cell is $O(r/n)$; and that part has length at least θr , since one side of x reaches at least that far within a cell of length $\geq r$. Hence the density is at least $\theta/2$ for every $\theta < 1$, so at least $\frac{1}{2}$.

Unrectifiability. This is the step that is elementary here and not in general. If a Lipschitz curve met Π in positive measure, Rademacher’s theorem and Mathlib’s partition of a differentiable map into pieces on which it approximates an invertible linear map would produce a set $A \subset [0, 1]$ of positive measure on which g is Lipschitz. But g jumps by at least α_n/n across every level- n boundary, so a set on which g is L -Lipschitz cannot have points within $\alpha_n/(2n(L+1))$ of a boundary on both sides. That alone does not make A null – a single such hole is a vanishing fraction of any ball – but a Lebesgue density point of A cannot lie that close to a level- n boundary for infinitely many n , because the ball of twice that radius about it would then carry a hole of relative size $\frac{1}{4}$. So almost every point of A eventually keeps its distance from the grid on *both* sides, and the set of points that do so at all fine levels is null by a nested recursion: inside each cell the surviving set is an interval, the grid points of the next level carve disjoint gaps out of it whose total is a fraction $1/(n(L+1))$ of its length, and $\sum 1/n = \infty$. No projection theorem, no tangent theory and no area formula are used.

7 What the machine checks

The certificate parameters were found by an unverified numerical search and are stored as exact rationals with denominator 10^4 . Nothing about how they were found enters the proof: Lean recomputes C , D , Q , B , ρ , m , β and U from the stored data by exact rational arithmetic and checks

$\alpha_t > 0$, $\eta_P, \eta_W \geq 0$ and $U \leq -1/2000$ for each of the thirty records (`gramCertificates_valid`). The positive semidefiniteness is a theorem, not a computation. The cover is checked by deciding, for each of the 64 ordered pairs of radius bands, that the band rectangle sits inside the rectangle of the assigned certificate.

stage	cost
thirty certificate validity checks	43 s
band cover	3 s
Gram certificate core	7 s

For comparison, the six-variable Bernstein formulation this replaces required an estimated 29 CPU-hours of kernel reduction over 3072 adaptive leaves, and was in any case incomplete: one of the ten coordinate charts had no certificate tree.

8 Provenance of the certificate data

The thirty certificates were checked three times independently before formalization: by an exact rational verifier, by a symbolic reconstruction using a computer algebra system, and by a third transcription written directly against the definitions above, which reproduced all thirty published values of U bit-for-bit. That agreement also validates the transcription into Lean, since a single mistyped integer would change U .